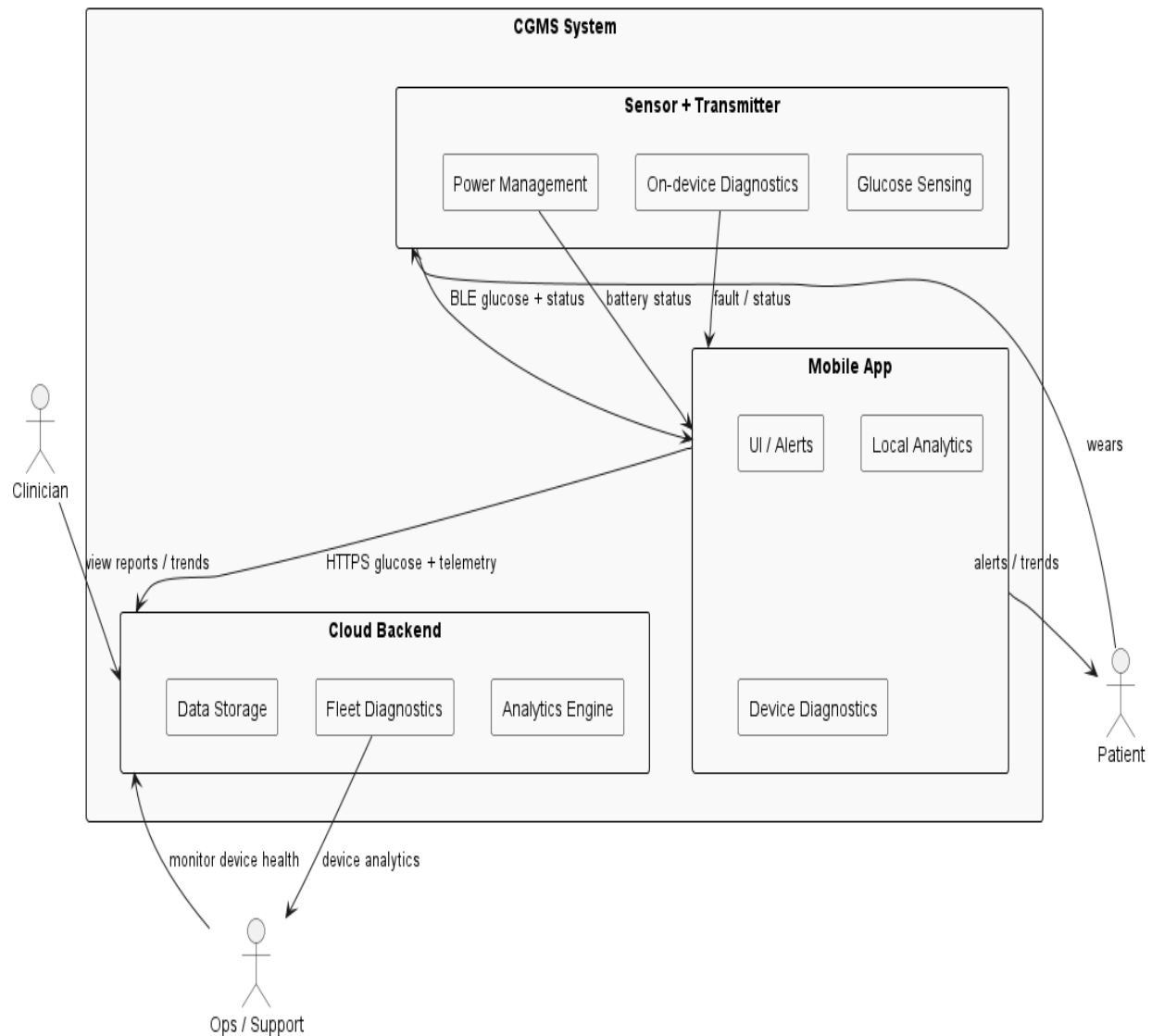


Continuous Glucose Monitoring System (CGMS)

System Architecture & Safety Design

1. System Context

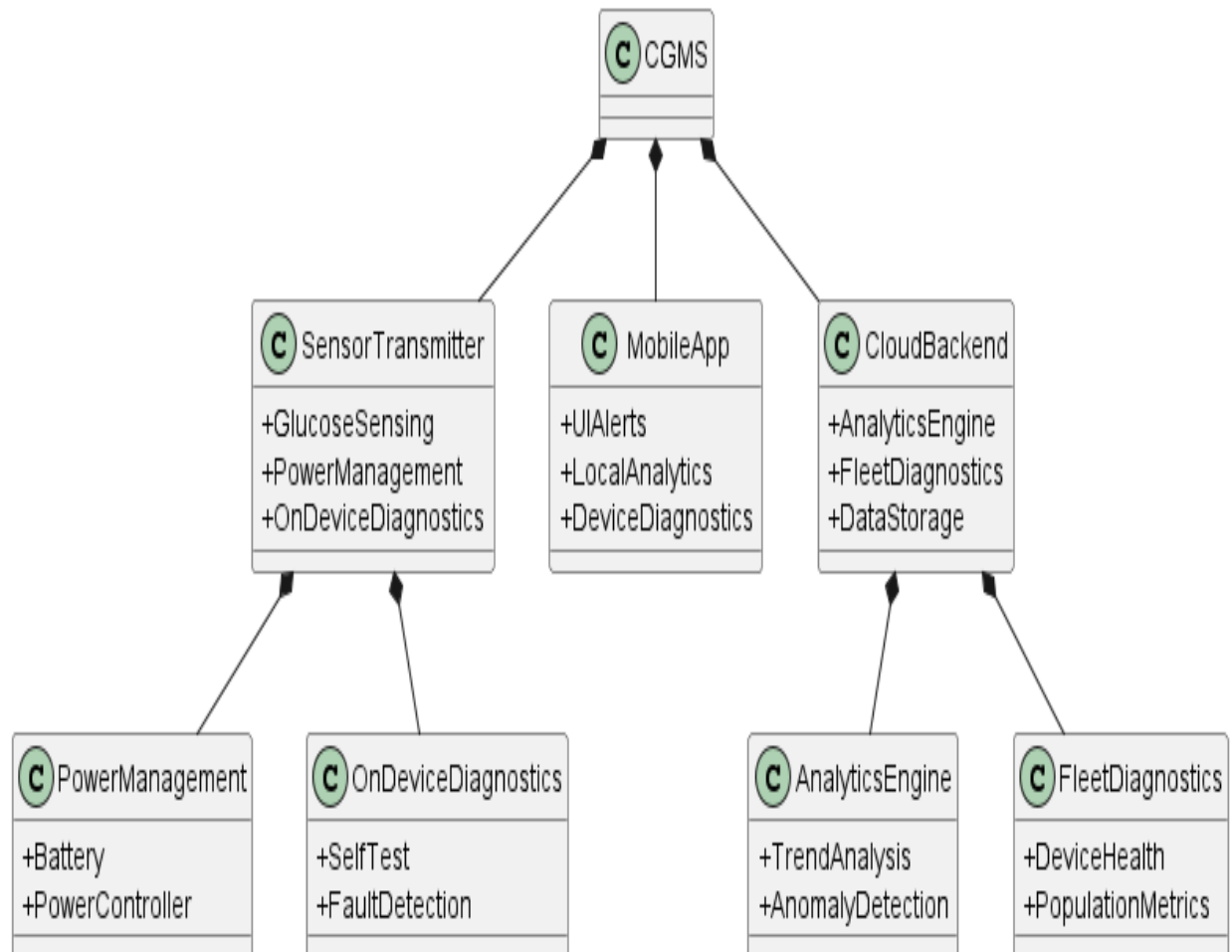


System Context

2. System Architecture (SysML)

2.1 Block Definition Diagram

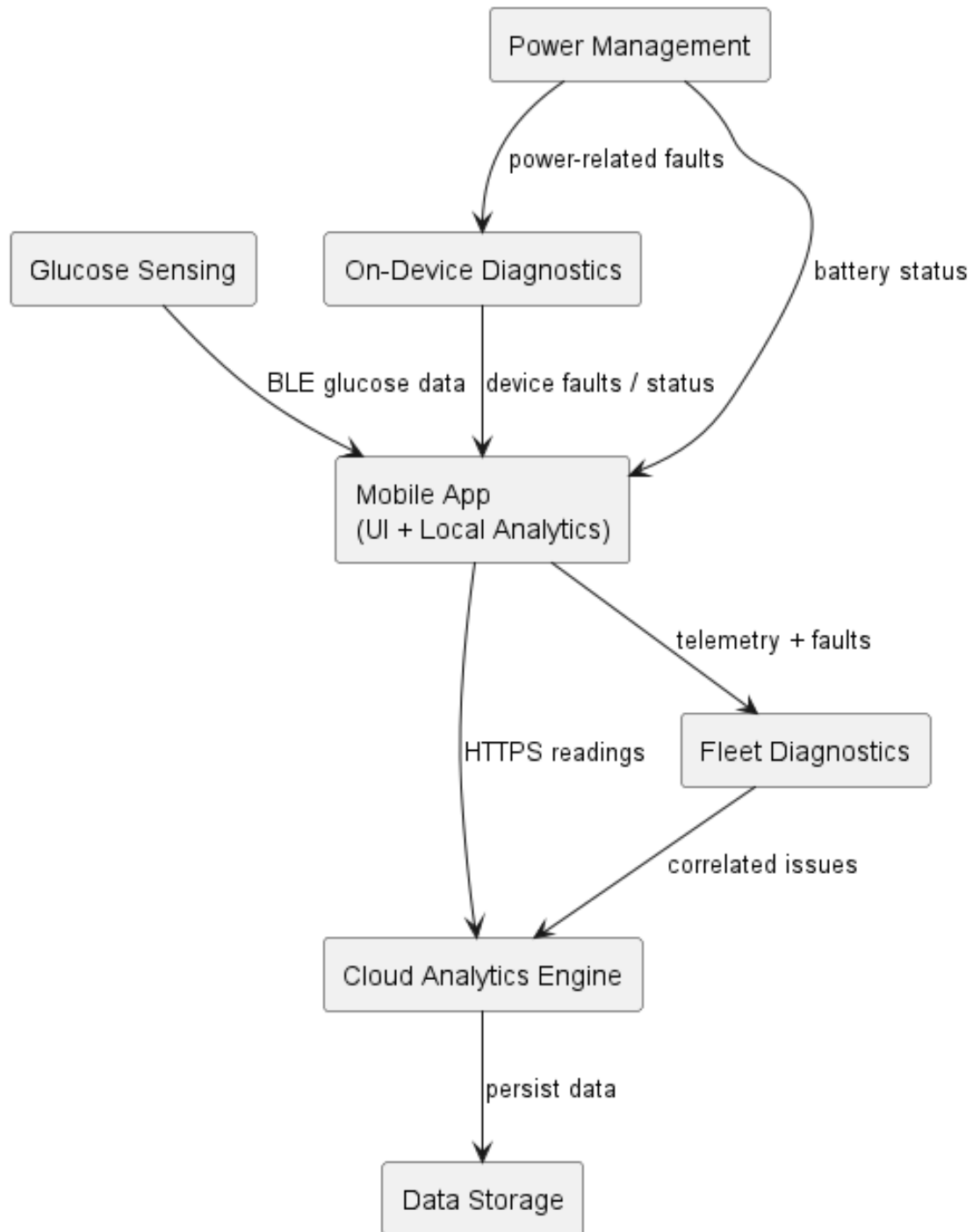
CGMS - Block Definition Diagram (BDD)



BDD

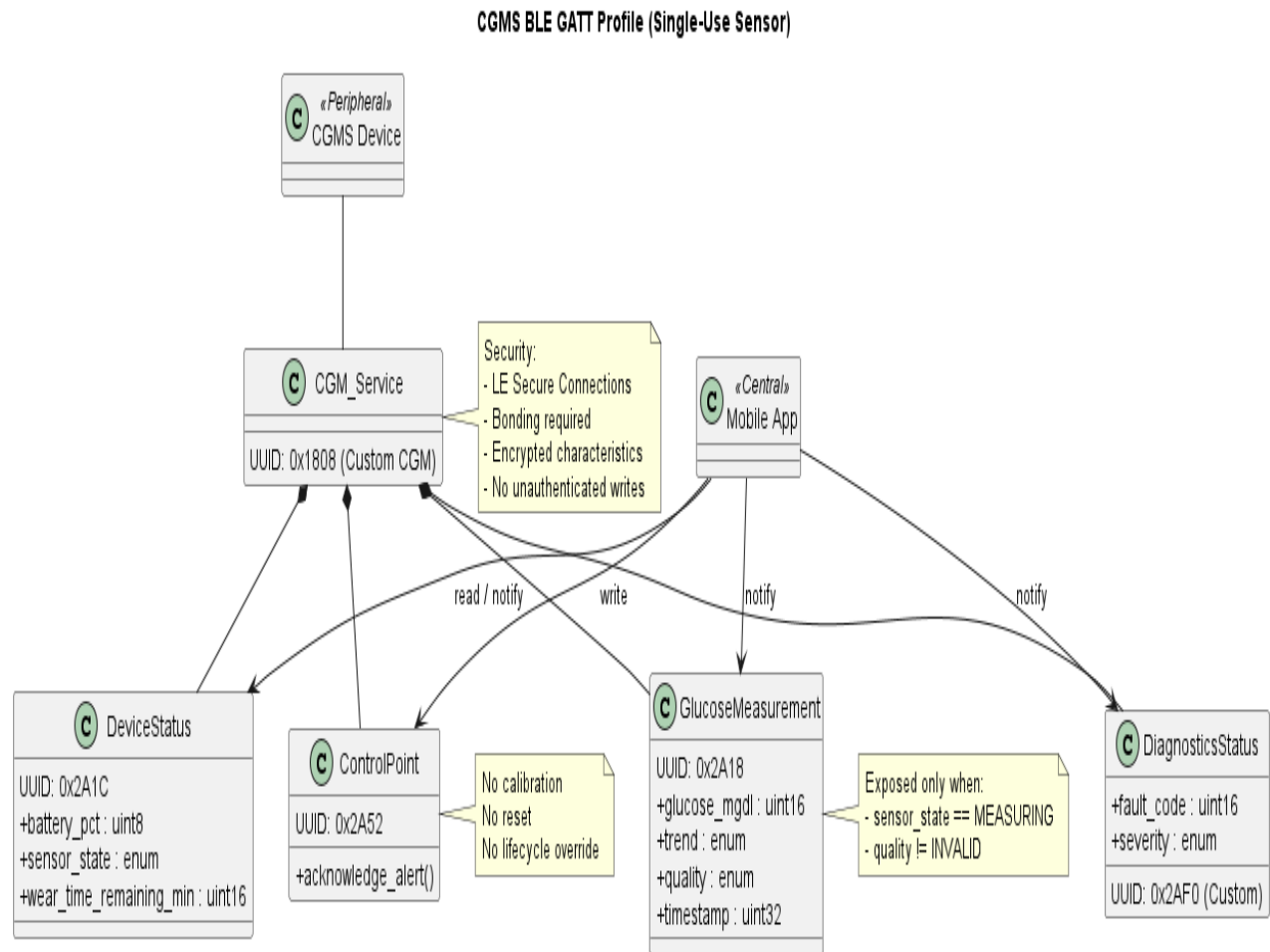
2.2 Internal Block Diagram

CGMS - Internal Block Diagram (IBD)



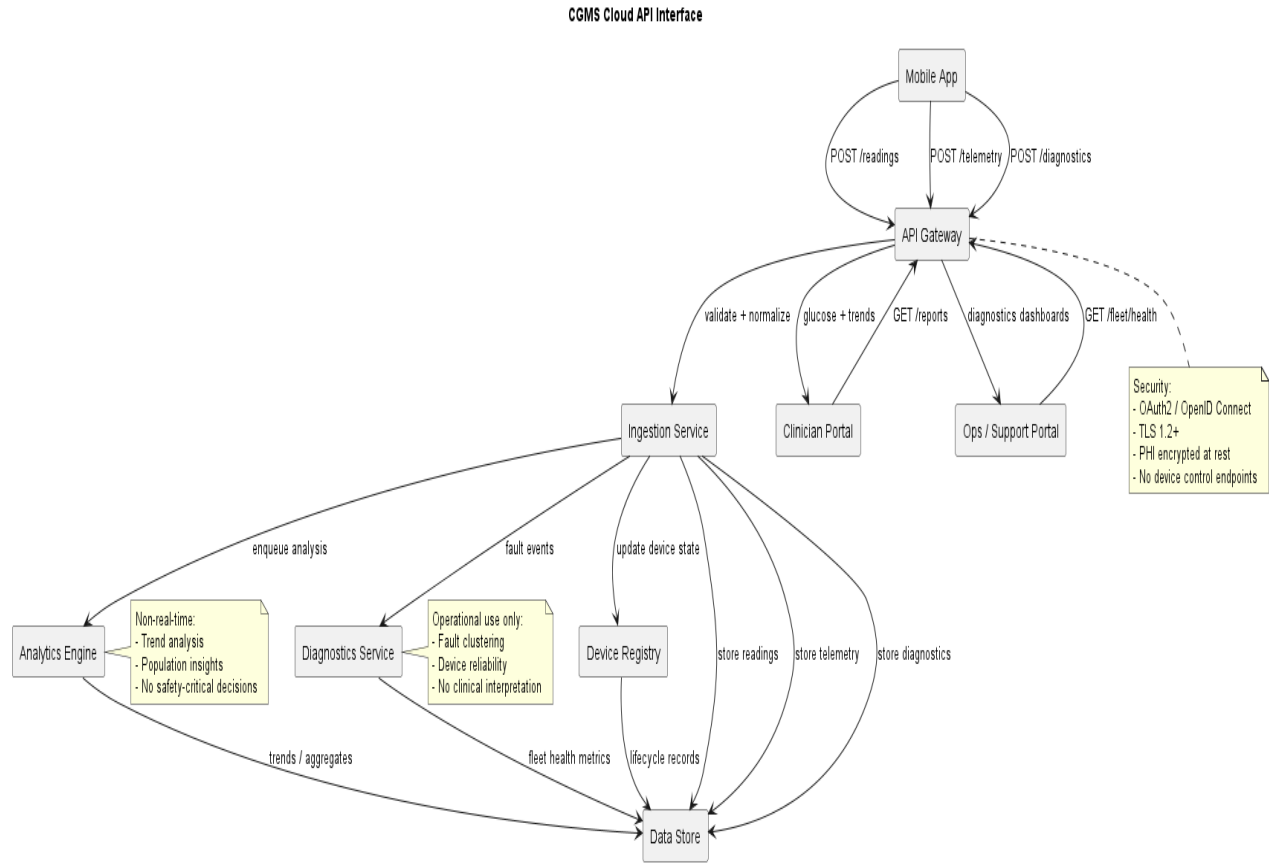
3. Interfaces

3.1 BLE GATT Interface



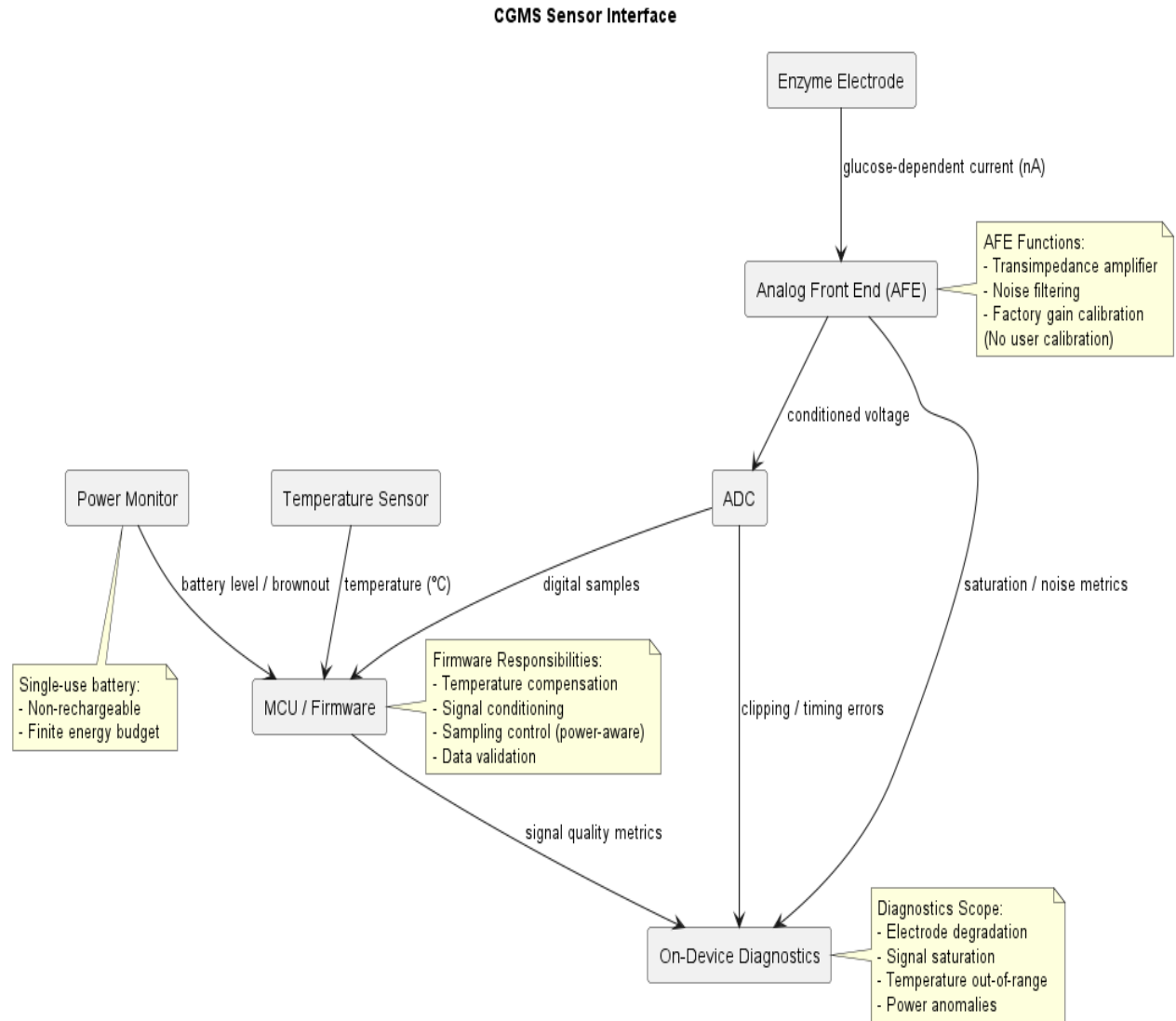
BLE GATT

3.2 Cloud API Interface



Cloud API

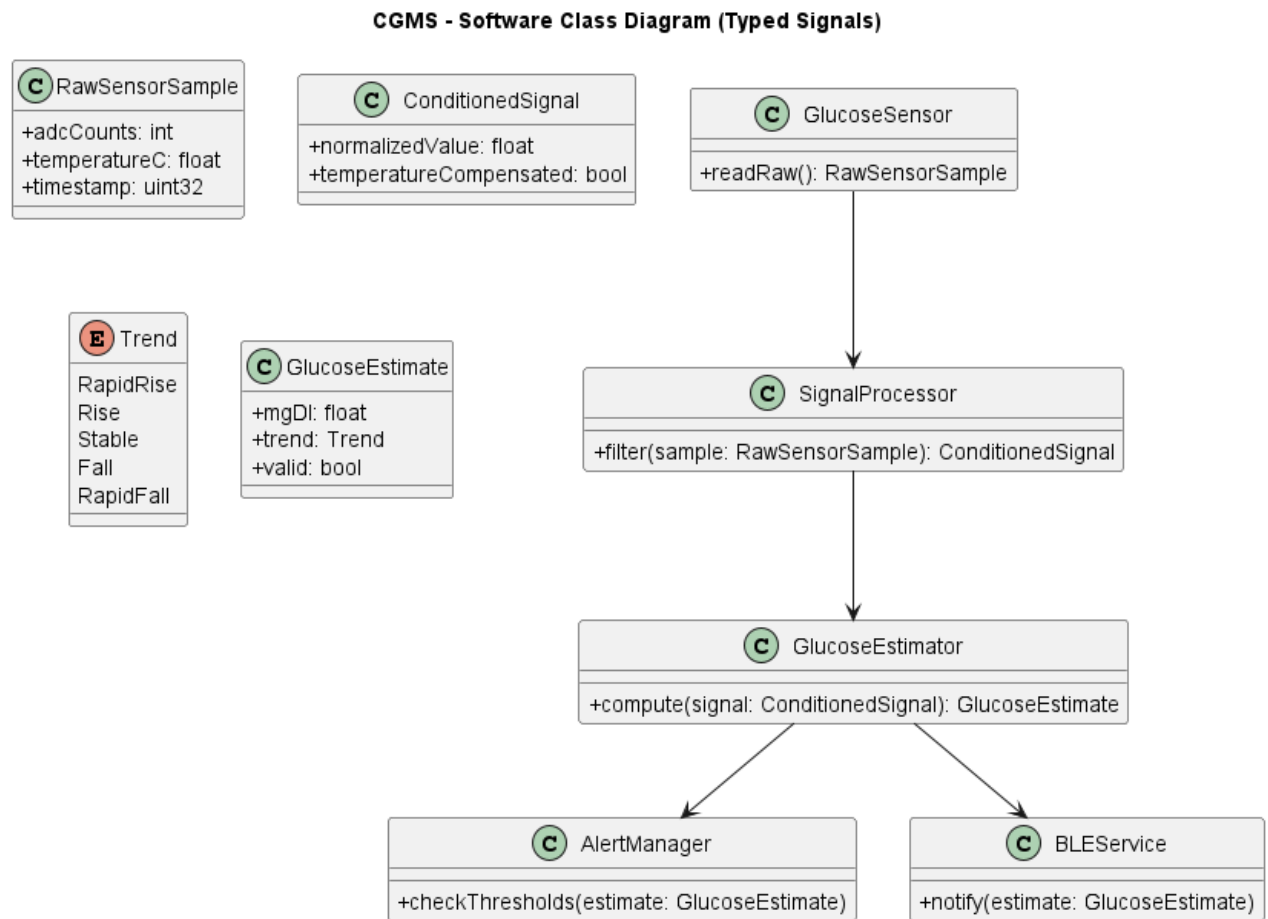
3.3 Sensor Interface



Sensor IF

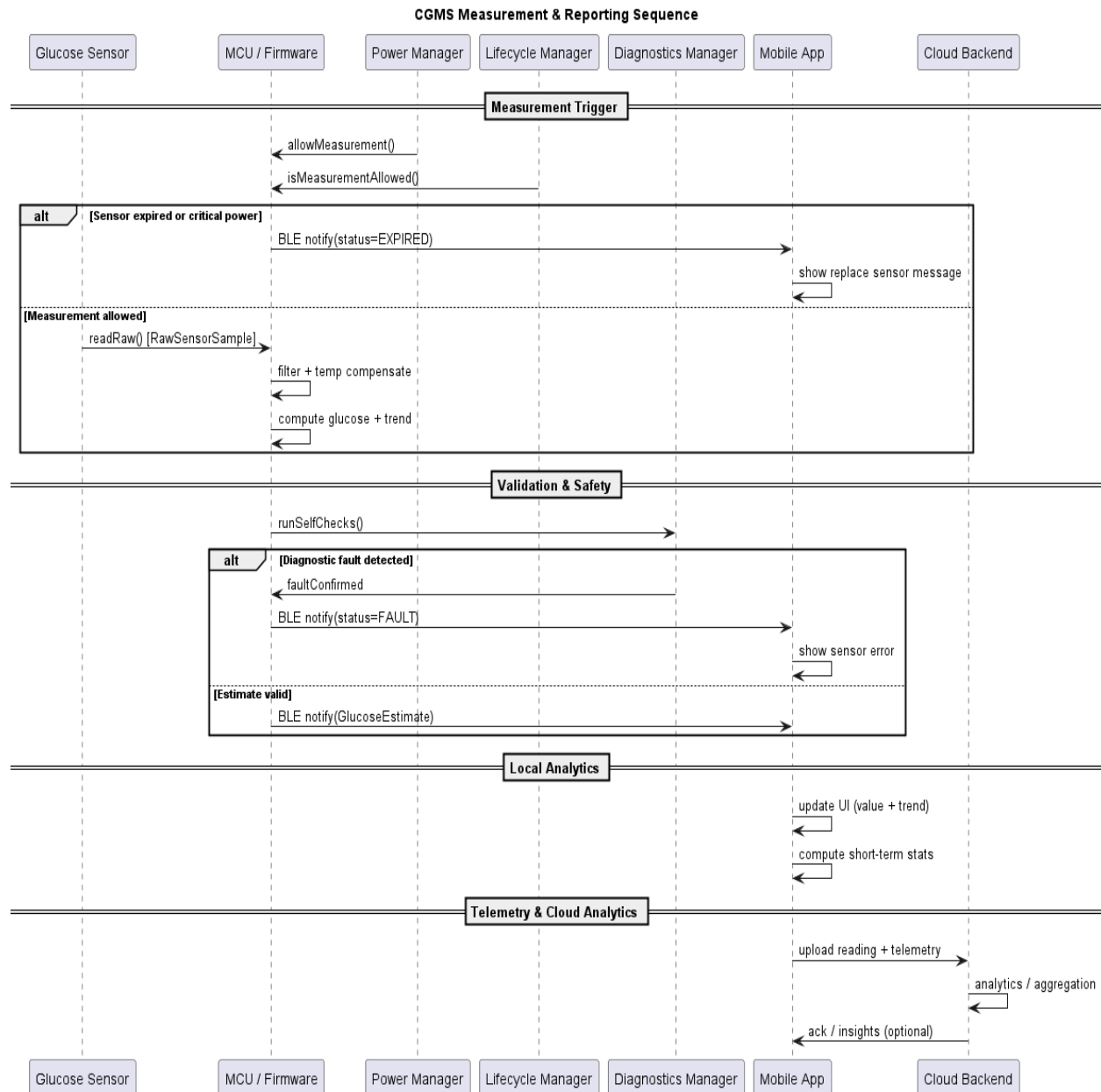
4. Software Design (UML)

4.1 Class Diagram



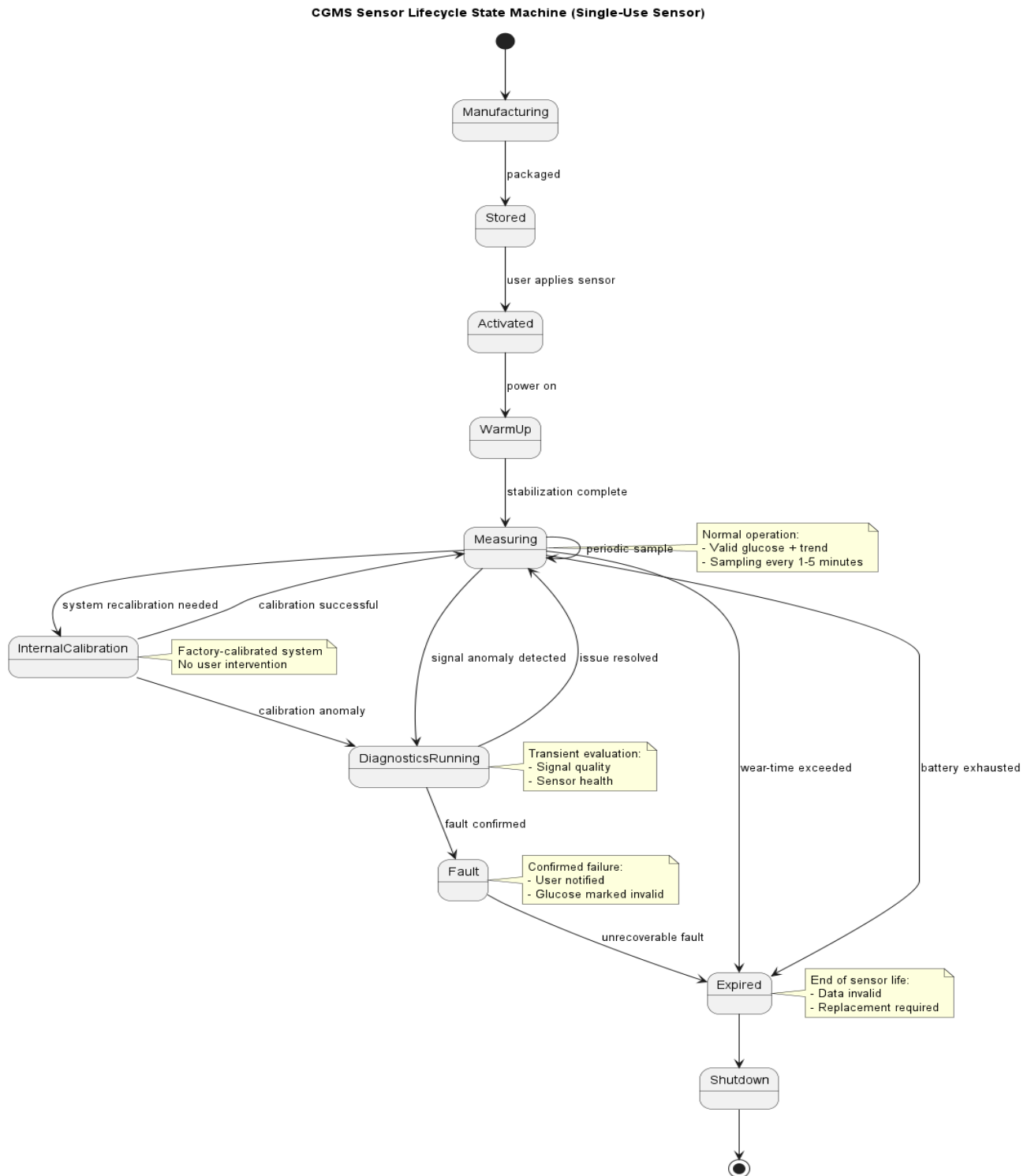
Class Diagram

4.2 Sequence Diagram



Sequence Diagram

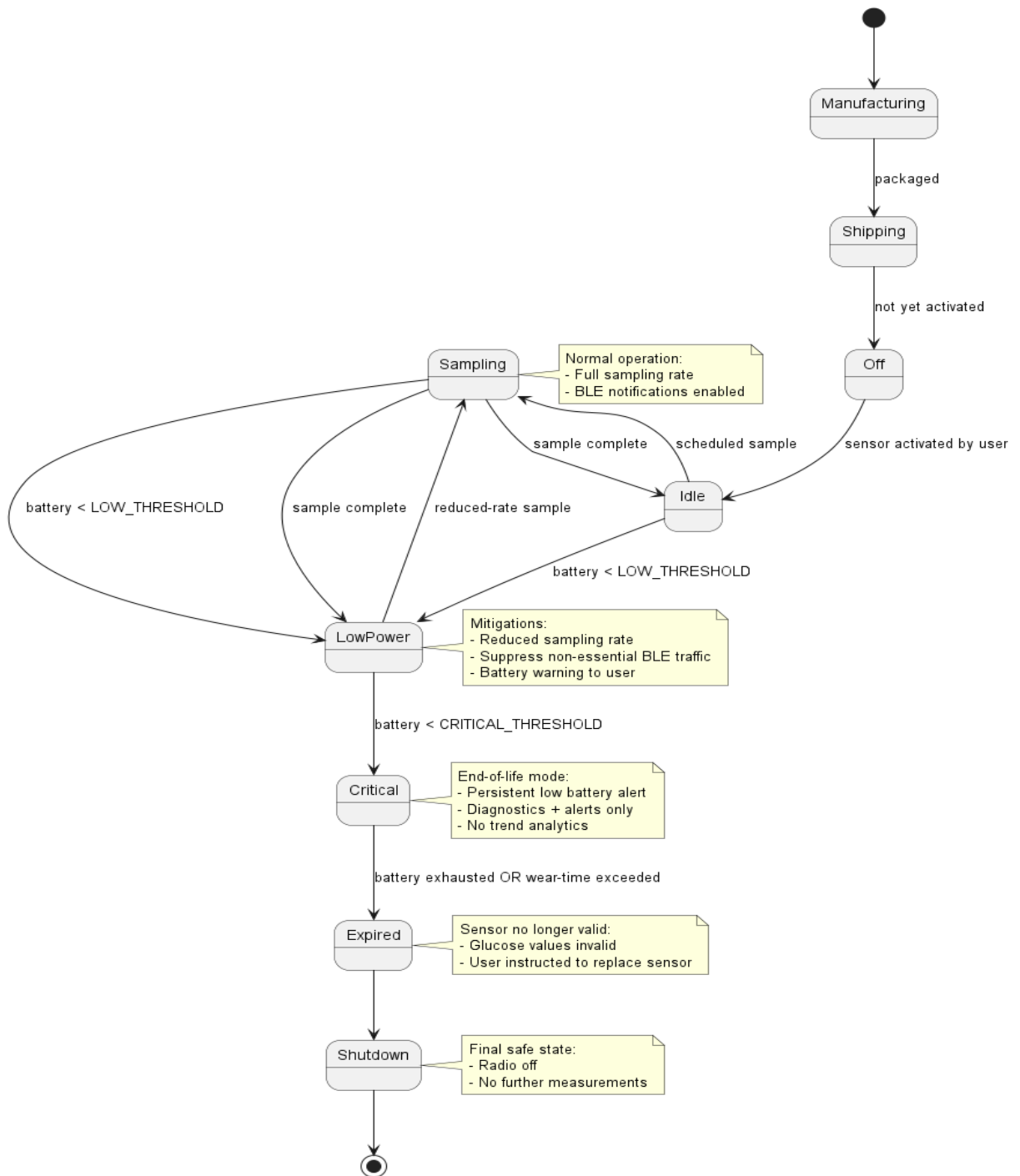
4.3 Sensor Lifecycle State Machine



Lifecycle State Machine

4.4 Sensor Power State Machine

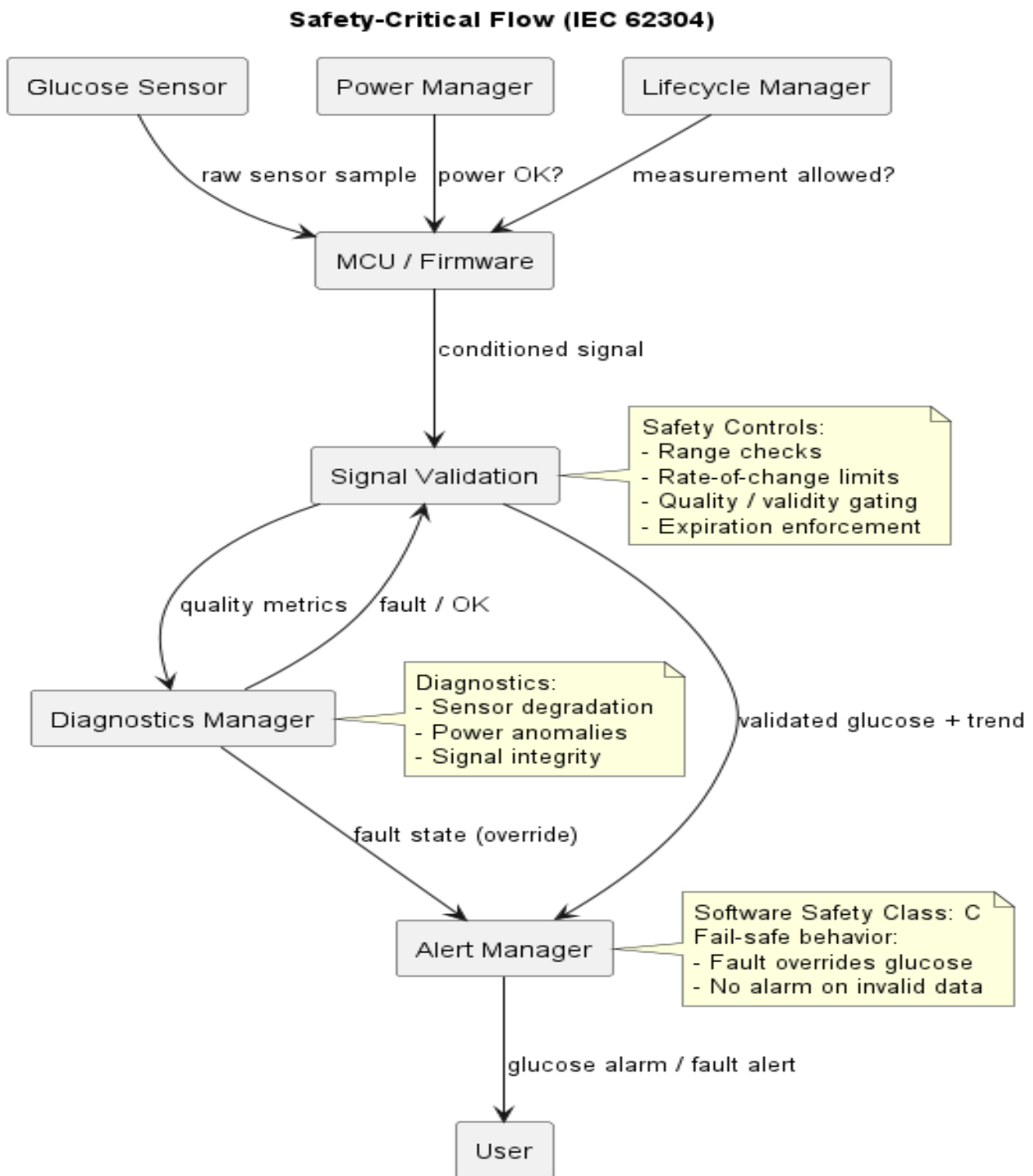
CGMS Power Management - Single-Use Sensor (Non-Rechargeable)



Power State Machine

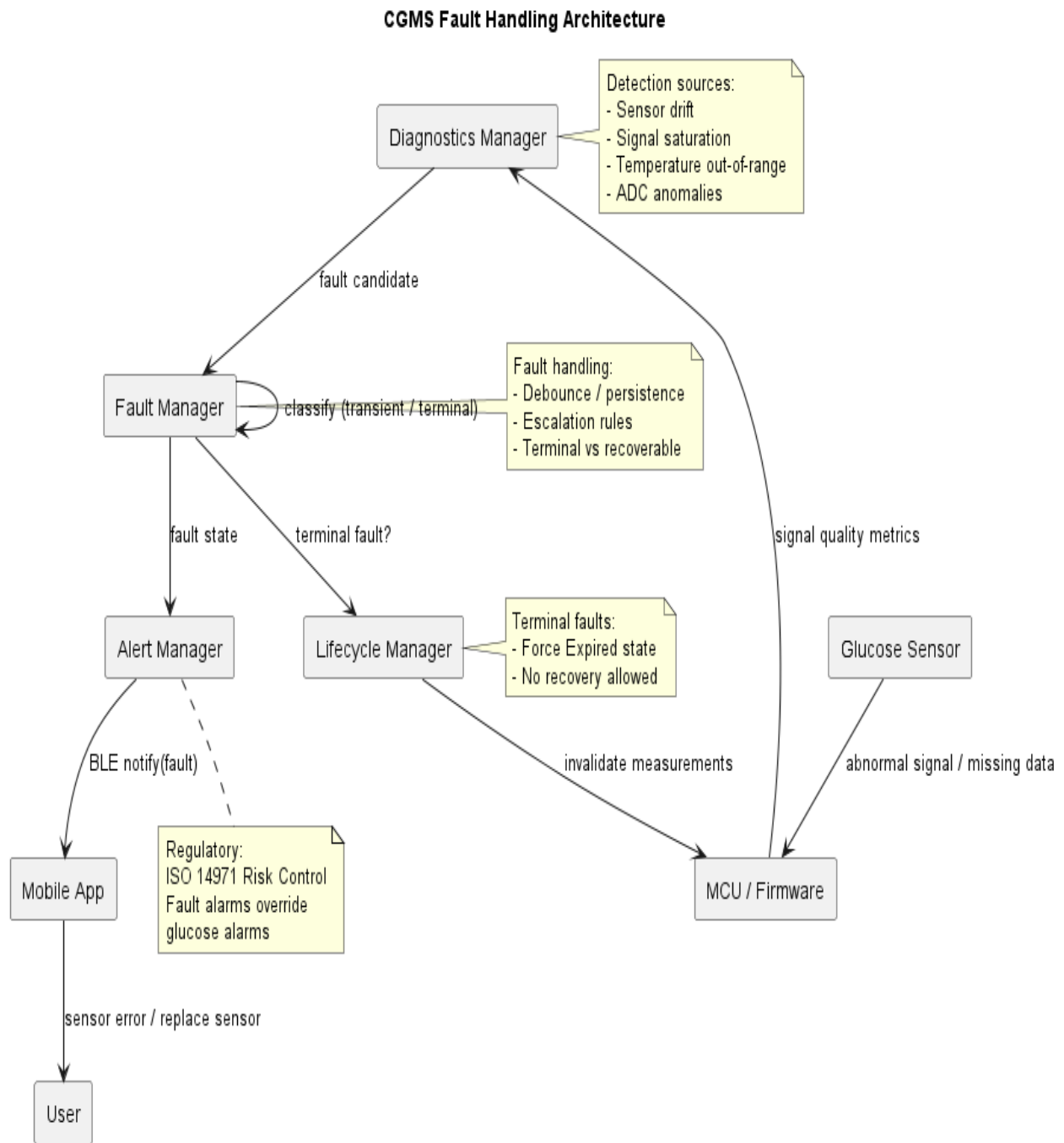
5. Safety & Risk Management

5.1 Safety-Critical Data Flow (IEC 62304)



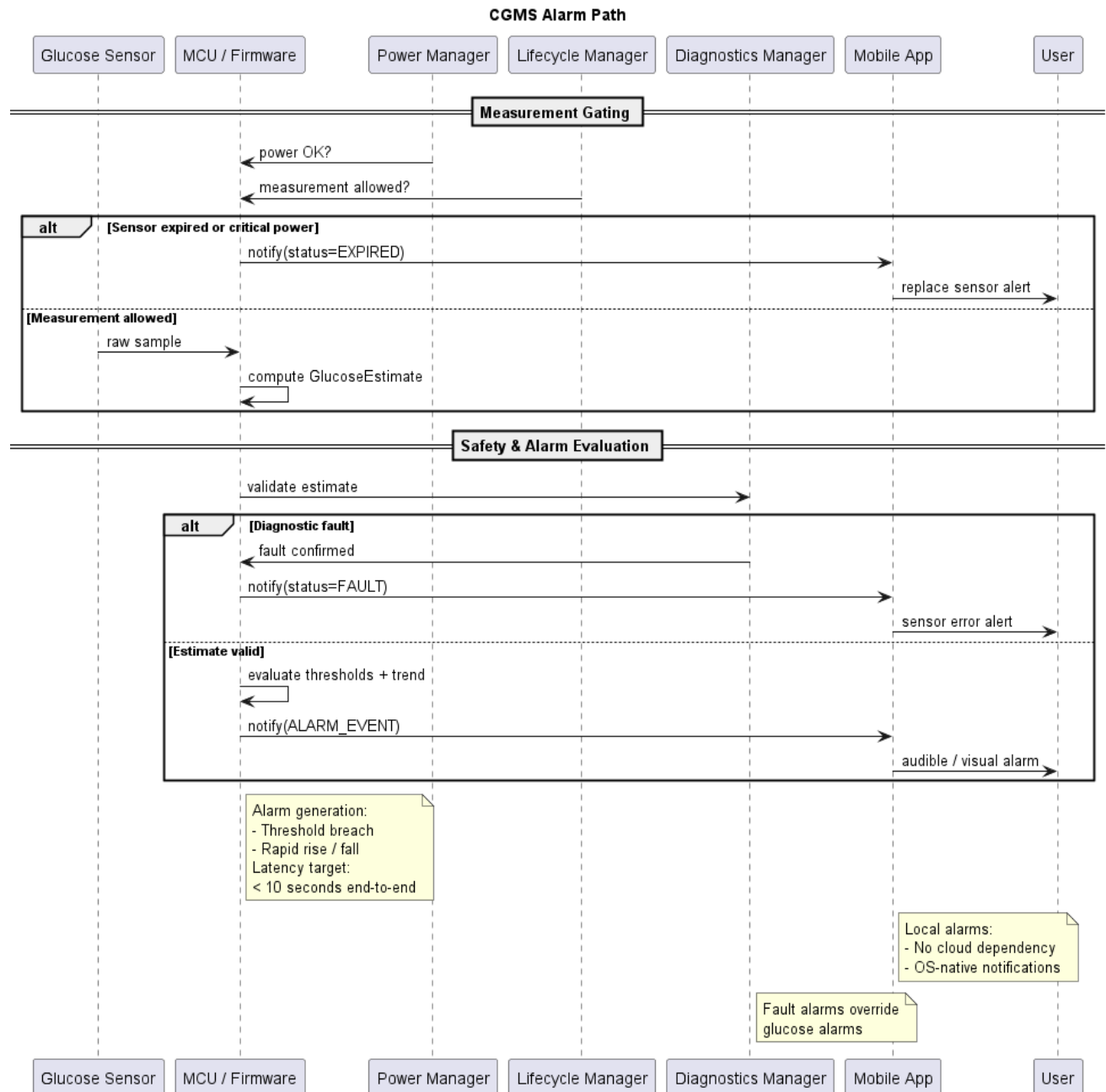
Safety Flow

5.2 Fault Handling



Fault Handling

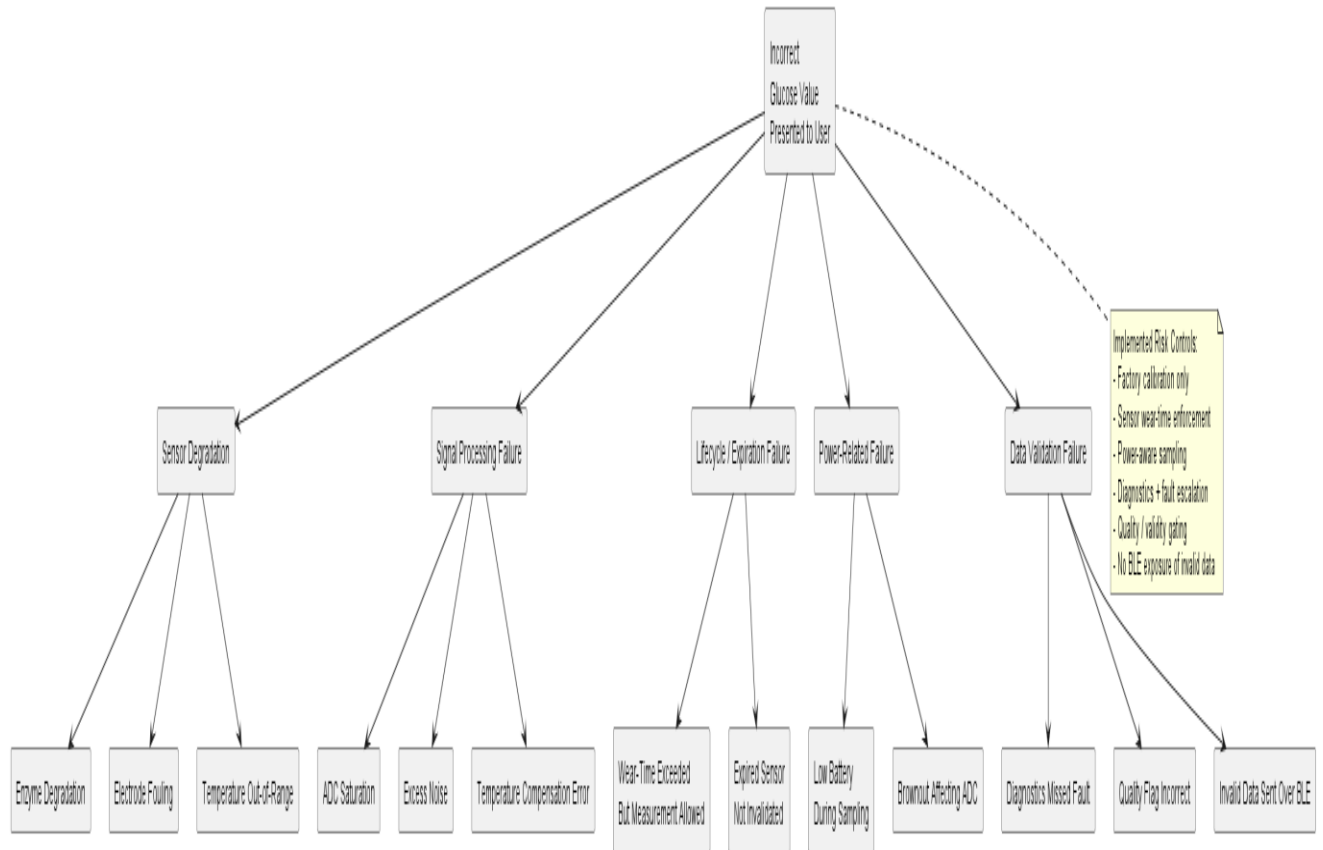
5.3 Alarm Path



Alarm Path

5.4 Fault Tree Analysis (ISO 14971)

Fault Tree Analysis - Incorrect Glucose Value



FTA